UNIVERSITAT POLITÈCNICA DE CATALUNYA
BARCELONATECH
Escola Superior d'Enginyeries Industrial,
Aeroespacial i Audiovisual de Terrassa

Assignment 1 on Reference Frames

Leoni Fritsch, Martí Güell, Gerard Martínez

Advanced Astrodynamics

ABSTRACT

This report computes the geocentric right ascension (α) and declination (δ) of Saturn near the Cassini "Grand Finale" (2017-09-15). The workflow propagates heliocentric orbital elements of Earth and Saturn to the observation epoch, forms state vectors in the perifocal frame, rotates them to the heliocentric ecliptic J2000 frame, converts to spherical ecliptic coordinates (λ, β), and finally transforms to the equatorial frame using the mean obliquity $\varepsilon(T)$. The geocentric relative vector Earth→Saturn is computed both to Saturn's center and to the planetary surface along the line of sight and shown to be practically identical in direction. The Kepler equation $M = E - e\sin E$ is evaluated with the numerical solver developed in *Assignment 2*.

1 METHOD

All computations use astronomical units (AU) and days unless stated otherwise. The Gaussian gravitational constant is $k = 0.01720209895$, hence the solar gravitational parameter in these units is $\mu_\odot = k^2$.

(1) Epoch: Julian Date and Julian Centuries

The UTC calendar date (Y, M, D) is converted to the Julian Date (JD) using the standard Gregorian algorithm. The epoch variable is the number of Julian centuries since J2000.0,

$$T = \frac{\mathrm{JD} - 2451545}{36525}.$$

This T consistently drives all time-dependent formulae (secular element rates, obliquity).

(2) Linear Propagation of Osculating Elements

Given reference elements X_0 and their secular rates $\dot{X}$ at J2000.0, the elements at epoch T are propagated as

$$X(T) = X_0 + \dot{X}\,T, \qquad X \in \{a, e, i, \Omega, \bar{\omega}, L\}.$$

Angles are reduced modulo 360°. The mean longitude L and the longitude of perihelion $\bar{\omega}$ satisfy $\bar{\omega} = \omega + \Omega$. (If an inclination appears negative in the source set, it is wrapped into $[0°, 180°]$ by $i \leftarrow 180° + i$ to preserve orientation.)

(3) Specific Angular Momentum

With semi–major axis a and eccentricity e, the specific angular momentum magnitude is

$$h = \sqrt{\mu_\odot\, a\,(1 - e^2)}.$$

This is later cross–checked numerically via $\|\mathbf{r} \times \mathbf{v}\| = h$.

(4) Mean and True Anomalies

The argument of perihelion and mean anomaly follow from

$$\omega = \bar{\omega} - \Omega, \qquad M = L - \bar{\omega}.$$

The eccentric anomaly E solves Kepler's equation $M = E - e\sin E$. **In this assignment, E is obtained with the numerical solver developed in Assignment 2.** The true anomaly θ is then recovered by

$$\cos\theta = \frac{\cos E - e}{1 - e\cos E}, \qquad \sin\theta = \frac{\sqrt{1-e^2}\,\sin E}{1 - e\cos E}.$$

(5) Perifocal State Vectors (PQW)

In the orbital plane (perifocal frame PQW),

$$r = \frac{h^2/\mu_\odot}{1 + e\cos\theta}, \quad \mathbf{r}_{pf} = r\begin{bmatrix}\cos\theta \\ \sin\theta \\ 0\end{bmatrix}, \quad \mathbf{v}_{pf} = \frac{\mu_\odot}{h}\begin{bmatrix}-\sin\theta \\ e + \cos\theta \\ 0\end{bmatrix}.$$

This yields heliocentric states for Earth and Saturn independently.

(6) Rotation to Heliocentric Ecliptic (J2000)

The inertial ecliptic frame (x-axis toward Aries) is obtained via the 3–1–3 rotation

$$\mathbf{r} = \mathbf{R}_3(\Omega)\,\mathbf{R}_1(i)\,\mathbf{R}_3(\omega)\,\mathbf{r}_{pf}, \qquad \mathbf{v} = \mathbf{R}_3(\Omega)\,\mathbf{R}_1(i)\,\mathbf{R}_3(\omega)\,\mathbf{v}_{pf}.$$

This places both planets in a common J2000 ecliptic frame suitable for differencing.

(7) Geocentric Relative Vector and Surface Point

The Earth→Saturn center vector is

$$\mathbf{r}_{ES} = \mathbf{r}_S - \mathbf{r}_E.$$

The nearest line-of-sight surface point is obtained by subtracting one Saturn radius along the sightline,

$$\mathbf{r}_{ES}^{\text{surf}} = \mathbf{r}_{ES} - R_S\,\hat{\mathbf{r}}_{ES}, \quad \hat{\mathbf{r}}_{ES} = \frac{\mathbf{r}_{ES}}{\|\mathbf{r}_{ES}\|}, \quad R_S = \frac{60268\ \text{km}}{149\,597\,870.7\ \text{km/AU}}.$$

Both vectors are practically collinear; they differ only by range ($\approx R_S$).

(8) Spherical Ecliptic Angles

Cartesian ecliptic coordinates (X, Y, Z) are mapped to spherical ecliptic coordinates by

$$\lambda = \text{atan2}(Y, X), \qquad \beta = \text{atan2}\Big(Z, \sqrt{X^2+Y^2}\Big), \qquad r = \sqrt{X^2+Y^2+Z^2},$$

with $\lambda \in [0, 360^\circ)$ and $\beta \in [-90^\circ, 90^\circ]$.

(9) Ecliptic to Equatorial (RA, Dec)

Using the mean obliquity $\varepsilon(T)$ (IAU 1980 polynomial; arcsec converted to degrees),

$$\begin{aligned}\sin\delta &= \sin\beta\cos\varepsilon + \cos\beta\sin\lambda\sin\varepsilon, \\ \cos\delta\cos\alpha &= \cos\beta\cos\lambda, \\ \cos\delta\sin\alpha &= -\sin\beta\sin\varepsilon + \cos\beta\sin\lambda\cos\varepsilon,\end{aligned}$$

then $\alpha = \text{atan2}(\cos\delta\sin\alpha, \cos\delta\cos\alpha)$ wrapped to $[0, 360^\circ)$ and optionally reported as hours via $\alpha_\text{h} = \alpha/15$.

2 INTERMEDIATE NUMERICAL RESULTS

Epoch used in the propagation: **2017-09-15** (UTC). Mean obliquity evaluated at **2017-11-15**; the difference with 2017-09-15 is $< 0.01^\circ$ and negligible at the present precision.

Julian quantities

JD	2458011.5
T (Julian centuries)	0.17704312114989734

Propagated orbital elements

Parameter	Earth	Saturn
a [AU]	1.000003605	9.536454530
e [–]	0.016703454	0.053771514
i [deg]	179.997693	2.486335
Ω [deg]	0.000000	113.611316
$\bar{\omega}$ [deg]	102.994915	92.524702
L [deg]	353.905830	266.388331

Derived quantities

Body	h (AU^2/day)	h (km^2/s)	ω (deg)	M (deg)
Earth	0.01719973004575414	4.455112879×10^9	102.99491530421109	250.91091430777726
Saturn	0.05304520705835192	$1.373988920 \times 10^{10}$	338.9133861346234	173.8636284853746

Table 1. Derived quantities and angular momentum in AU units and converted to km units.

The corresponding true anomalies printed by the solver were

$$\theta_{\mathrm{Earth}} = 4.347870977300497 \text{ rad}, \qquad \theta_{\mathrm{Saturn}} = 3.045266631551506 \text{ rad}.$$

Heliocentric ecliptic state vectors

The Python output provided the state vectors both in AU/AU/day and in km/km/s:

Body	**r** (AU)	**v** (AU/day)
Earth	$(0.996191705,\ 0.138063392,\ -5.5601\times10^{-6})$	$(2.08179888\times10^{-3},\ -1.69769631\times10^{-2},\ 6.83700704\times10^{-7})$
Saturn	$(-0.52093175,\ -10.03118403,\ 0.19518666)$	$(5.26714\times10^{-3},\ -3.0640\times10^{-4},\ -2.0423\times10^{-4})$

Table 2. Heliocentric ecliptic states in canonical units.

Body	**r** (km)	**v** (km/s)
Earth	$(1.49028158\times10^{8},\ 2.06539894\times10^{7},\ -8.31782872\times10^{2})$	$(3.60454490,\ -29.3948788,\ 1.18379826\times10^{-3})$
Saturn	$(-7.79302799\times10^{7},\ -1.50064377\times10^{9},\ 2.91995093\times10^{7})$	$(9.11982685,\ -0.53051334,\ -0.35362264)$

Table 3. Same states converted to km and km/s using 1 AU = 149 597 870.7 km.

Geocentric relative vector and surface comparison

Quantity	Value (AU)	Comment
$\mathbf{r}_{ES}$	$(-1.51712345,\ -10.16924743,\ 0.19519222)$	Earth→Saturn center
$\|\mathbf{r}_{ES}\|$	10.283645112	center distance
$\mathbf{r}_{ES}^{\mathrm{surf}}$	$(-1.51706402,\ -10.16884904,\ 0.19518458)$	line-of-sight surface point
$\|\mathbf{r}_{ES}^{\mathrm{surf}}\|$	10.283242245	surface distance
$\Delta\|\mathbf{r}\|$	0.000402867	$\approx R_S$ (6.0268×10^4 km)
Rel. difference	3.917548×10^{-5}	directions coincide (angle ≈ 0)

Table 4. Earth–Saturn relative vectors exactly as printed by the Python script.

Ecliptic and equatorial coordinates of Earth→Saturn

Coordinates	λ (deg)	β (deg)	r (AU)
Ecliptic (geocentric)	261.514775	1.087587	10.283645112

Table 5. Geocentric ecliptic coordinates of Saturn on 2017-09-15 from Python output.

Mean obliquity at 2017-11-15: $\varepsilon = 23.436967^\circ$.

Quantity	Value (deg)	Comment
Right ascension α	260.839315	17.389288 h
Declination δ	-22.079980	southern sky

Table 6. Equatorial coordinates of Saturn (geocentric) obtained after ecliptic–equatorial rotation.

3 CONCLUSION

Using propagated heliocentric elements and standard frame rotations, the geocentric ecliptic coordinates of Saturn on 2017-09-15 are

$$(\lambda, \beta, r) = (261.514775^\circ,\ 1.087587^\circ,\ 10.283645112\ \mathrm{AU}),$$

which corresponds to

$$r = 1.538411412 \times 10^9\ \mathrm{km}$$

when converted with 1 AU = 149 597 870.7 km. Transformation with the mean obliquity yields equatorial coordinates

$$\alpha = 260.839315^\circ\ (17^{\mathrm{h}}\,23^{\mathrm{m}}\,21.4^{\mathrm{s}}), \qquad \delta = -22.079980^\circ,$$

consistent with Saturn's southern-sky position during Cassini's end-of-mission period. The Earth→Saturn center and surface vectors are indistinguishable in direction, differing in magnitude only by one Saturn radius, so the center vector is used for all angle computations. The additional km- and km/s-level outputs produced by the Python script are now explicitly documented in the method and numerical results sections.